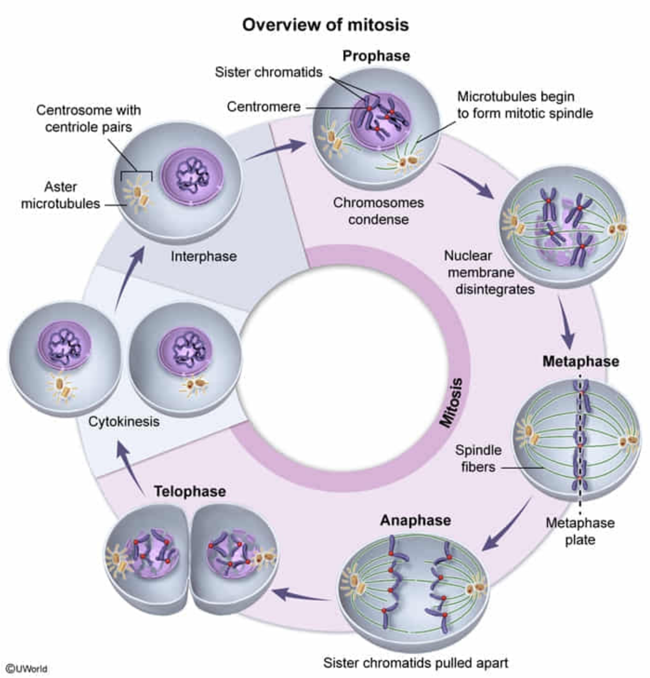
Researchers are studying the last two phases of mitosis, anaphase and telophase, in actively dividing cancer cells. Different fluorescent probes are used to label various cellular components in these cells. Under the microscope, fluorescently labeled DNA appears blue, the nuclear envelope appears green, and the mitotic spindle appears red. Given this, green fluorescence would be most intense during:

- A. anaphase only. [5%]
- B. telophase only. [78%]
- C. both anaphase and telophase. [4%]
- D. neither anaphase nor telophase. [11%]

Correct
77% answered correctly

Explanation:



Following cell growth and DNA replication during interphase, somatic (non-sex) eukaryotic cells divide via mitosis, producing two genetically identical daughter cells. Mitosis includes the following phases:

1. **Prophase**: DNA condenses to form chromatids. Each pair of sister (identical) chromatids are joined by a region called the centromere to form chromosomes. The nuclear envelope breaks down and centrosomes (microtubule-organizing structures) migrate to opposite poles within the cell. The mitotic spindle is formed as microtubules grow from these

centrosomes.

2. **Metaphase**: Chromosomes attach to spindle fiber microtubules at their kinetochores and align at the metaphase plate, a central plane within the cell.

3. **Anaphase**: Sister chromatids are pulled apart by the spindle fibers and move toward opposite poles of the cell. This forms two sets of chromosomes within the cell (one set at each cellular pole).

4. **Telophase**: The nuclear envelope is reformed around each set of chromosomes. Chromosomes decondense and the parental cell undergoes cytokinesis (cytoplasmic division) to produce two identical daughter cells.

In the given scenario, researchers fluorescently labeled various components of actively dividing cancer cells and visualized them under a microscope. Specifically, the nuclear envelope appeared green when viewed by the researchers. In mitosis, the **nuclear envelope** breaks down during prophase and **reforms during telophase**. Therefore, green fluorescence would be **most intense during telophase** as the nuclear envelope reforms.

(Choices A, C, and D) The nuclear envelope *breaks down* during prophase, allowing progression of metaphase and anaphase in the cytosol. Because the nuclear envelope has *not yet* reformed, green fluorescent staining of the nuclear envelope would be *least* intense during both metaphase and *anaphase*. In contrast, green fluorescent staining of the nuclear envelope would be *most* intense during reformation of the nuclear envelope in telophase.

Educational objective:
Mitosis in eukaryotic cells includes four distinct phases: prophase, metaphase, anaphase, and telophase. The nuclear envelope breaks down during prophase, allowing both metaphase and anaphase to occur in the cytoplasm. During telophase, the nuclear envelope reforms around the chromosomes prior to cytokinesis, the cytoplasmic division of the parental cell into two identical daughter cells.

Time spent:0 sec
Subject:
Biology

QID:402409
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